Radius - Scheduling

Ticket: WEB-1812: TRACKING - Timezone-Aware Hierarchical Scheduling EngineTo Do

1. Executive Summary

The Radius ad platform is introducing a smart, timezone-aware scheduling engine that provides consistent control across Campaigns, Line Items, and Creatives. While most industry-standard ad systems already support scheduling across these three tiers, the current Radius platform relies on manual status changes that create operational friction when dates expire or campaigns need extensions. This update brings the platform in line with broader industry standards by enabling real-time schedule evaluation rather than locking objects into inactive statuses. This change ensures precise delivery control across global time zones, simplifies the process of extending campaign flights, and allows for flexible, automated management of promotions and creative assets without requiring manual intervention.

Business Benefits

- **Revenue Protection:** Eliminates non-billable over-delivery by dynamically suppressing serving the instant any parent or child schedule expires.
- **Global Operational Scalability:** Empowers global AdOps teams to execute localized campaigns across timezones using explicit IANA designations.
- **Workflow Efficiency:** Reduces manual operational overhead via "Immediate" start triggers and "Indefinite" end-date flags.
- **Flighting Flexibility:** Allows buyers to extend expired flights or update end dates without requiring manual entity reactivation or status resets.
- **Granular Asset Management:** Enables traders to schedule promo-specific or time-gated Creatives within continuously active Line Items.

2. Engineering Gist

The core objective is to shift from static status transitions to a dynamic runtime evaluation model within the Radius ad decision pipeline. Schedule validity must be evaluated as a computed runtime state against _T_now rather than permanently altering an entity's persistent administrative status (e.g., setting status = INACTIVE).

- **Data Normalization:** Establish UTC timestamp normalization across Campaigns, Line Items, and Creatives, paired with explicit IANA timezone metadata for UI presentation

and reporting reconstruction.

- **Session Lifecycle Management:** During ad request processing, the ad server evaluates schedule eligibility sequentially across the target hierarchy: Campaign → Line Item → Creative. An entity is eligible if and only if _T_start ≤ _T_now < _T_end. If an end datetime is updated into the future, subsequent ad evaluations dynamically recognize the entity as eligible without requiring state resets.
- **Downstream Data Availability:** Real-time computed evaluation states (e.g., active vs. schedule-expired) must be streamed asynchronously to reporting storage without adding blocking overhead to the ad decision path.
- **Data Isolation:** Scheduling rules and evaluation event logs must be isolated from core transactional serving tables to prevent table locks and schema collisions.

3. Detailed Specifications

Use Cases and User Stories

- **Internal AdOps Specialist:** "As an AdOps Specialist, I need to configure start and end datetimes with explicit timezones on individual Creatives so that promotional video overlays automatically run only within specific promotional windows."
- **External DSP Trader / Media Buyer:** "As a Media Buyer, I need to extend the end date of a Line Item without having to manually recreate or reactivate downstream Creatives that were suppressed purely due to the previous schedule expiring."
- **Data Analyst:** "As a Data Analyst, I need normalized UTC timestamps and localized timezone attributes recorded in ad delivery logs to accurately reconcile impression delivery across regional flight times."

Requirements

Object Configuration and Scheduling Capabilities

1. The **Radius UI MUST** present required start datetime, required end datetime, and explicit IANA timezone selection controls across Campaign, Line Item, and Creative management interfaces. The timezone control **MUST** default to America/New_York, the start datetime **MUST** default to “Immediate”, and the end datetime **MUST** default to “Indefinite”.
2. The **Radius System MUST** allow users to select an "Immediate" start flag for Campaigns, Line Items, and Creatives, setting the effective start timestamp to the exact time of object save or activation.
3. The **Radius System MUST** allow users to select an "Indefinite" end flag for Campaigns, Line Items, and Creatives, bypassing end-date expiration evaluation for that entity.

4. The **Radius Database MUST** store all scheduling inputs as normalized UTC timestamps alongside the corresponding IANA timezone identifier string (e.g., America/New_York).

HubSpot Integration Requirements

5. For new Campaign creation initiated by a HubSpot push, the **Radius System MUST** set the Campaign start datetime, end datetime, and timezone in Radius using the scheduling values supplied by HubSpot at creation time only.
6. The **Radius System MUST NOT** update a Campaign's start datetime, end datetime, or timezone in response to subsequent edits made in HubSpot after the Campaign has been created in Radius.
7. The **Radius UI MUST** allow users to edit the start datetime and end datetime for Campaigns created from HubSpot after initial creation, with Radius remaining the system of record for those scheduling fields after creation.

Dynamic Hierarchical Evaluation Logic

8. The **Ad Server MUST** dynamically evaluate scheduling eligibility for every ad request by comparing current UTC time (_T_now) against the persisted UTC schedule parameters.
9. The **Ad Server MUST** immediately halt ad delivery for a Creative if ANY entity in its hierarchy (Campaign, Line Item, or Creative) has an effective end datetime where _T_now ≥ _T_end.
10. The **Ad Server MUST NOT** serve a Creative prior to its effective start datetime (_T_now < _T_start) or after its effective end datetime (_T_now ≥ _T_end).
11. The **Radius System MUST NOT** mutate an object's administrative status field (e.g., setting status to DISABLED or INACTIVE) strictly because an end datetime has passed, ensuring schedule eligibility remains purely dynamic.
12. The **Radius System MUST** automatically resume ad serving eligibility for an entity if its end datetime is modified to a future timestamp, provided its overarching administrative status remains active.
13. The **Radius UI MUST** display computed runtime availability states (e.g., "Active", "Pending Start", "Schedule Expired", "Parent Expired") without altering underlying administrative status configurations.

Data Warehouse and Storage Isolation

14. The **Data Warehouse MUST** isolate campaign scheduling metadata and evaluation state audit logs within dedicated namespaces to prevent schema collisions with primary transactional ad-serving tables.

15. The **Ad Server MUST** asynchronously stream dynamic evaluation events to downstream reporting storage without blocking the real-time ad decision loop.

Migration Requirements

16. If objects currently lack start or end date configurations, no date parameters shall be added during the migration process.
17. For objects that currently possess start or end dates and are actively serving, the migration process must identify these objects for the release without modifying or adding new constraints that would disrupt their current serving status.

4. Workflow Logic

1. **Schedule Configuration (UI Ingress):**

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    - The UI defaults scheduling inputs to Start Datetime = “Immediate”, End Datetime = “Indefinite”, and IANA Timezone = America/New_York for a Campaign, Line Item, or Creative.
        - The user may change any of these defaults by entering a specific Start Datetime, entering a specific End Datetime, and/or selecting a different IANA Timezone before saving.
        - The user must be able to select “Immediate” and “Indefinite” for start and end at any time.

2. **Normalization & Persistence:**
    - Radius System converts input datetimes from the specified timezone into normalized UTC timestamps.
    - Radius Database persists normalized UTC timestamps, IANA timezone string, and administrative status (Active/Inactive).
3. **Real-Time Ad Request Ingress (*T*now UTC):**
    - Ad Server receives request and retrieves target hierarchy: Campaign -> Line Item -> Creative.
4. **Hierarchical Dynamic Evaluation:**
    - **Campaign Evaluation:** Check if Administrative Status is Active AND _T_start ≤ _T_now AND (_T_end is Indefinite OR _T_now < _T_end).
        - If *FALSE*: Stop processing. Exclude hierarchy from delivery (Computed State: Campaign Expired/Pending).
        - If *TRUE*: Proceed to Line Item evaluation.
    - **Line Item Evaluation:** Check if Administrative Status is Active AND _T_start ≤ _T_now AND (_T_end is Indefinite OR _T_now < _T_end).

    - If *FALSE*: Stop processing. Exclude hierarchy from delivery (Computed State: Line Item Expired/Pending).
    - If *TRUE*: Proceed to Creative evaluation.
  - **Creative Evaluation:** Check if Administrative Status is Active AND _T_start ≤ _T_now AND (_T_end is Indefinite OR _T_now < _T_end).
    - If *FALSE*: Exclude Creative from delivery (Computed State: Creative Expired/Pending).
    - If *TRUE*: Mark Creative as eligible for auction/serving.
5. **Ad Delivery & Asynchronous Logging:**
   - Ad Server returns eligible Creative payload (VAST/SIMID).
   - Evaluation decision details are pushed asynchronously to the isolated Data Warehouse namespace.
6. **Schedule Modification Flow:**
   - User updates Tend on an expired Line Item to a future date in the UI.
   - Database updates the stored UTC timestamp. Administrative status remains Active.
   - On the subsequent ad request, step 4 evaluates *T_now < _T_end as _TRUE*, immediately resuming delivery without manual status intervention.

5. Terminology and References

- **Radius:** KERV's core proprietary ad server responsible for ad decisioning, campaign management, and interactive overlay generation.
- **Line Item:** An intermediate programmatic entity within Radius linking Creatives to budgets, targeting parameters, and flight schedules under a parent Campaign.
- **IANA Timezone Database:** The standard, internationally recognized repository of global time zones and daylight saving time rules.
- **VAST (Video Ad Serving Template):** The IAB standard XML specification for video ad structures.
- **SIMID (Secure Interactive Media Interface Definition):** The IAB standard allowing interactive overlays to execute securely within video player environments.
- **RFC 2119:** Official standards document defining key requirement terms (**MUST**, **SHOULD**, **MUST NOT**, **SHOULD NOT**).